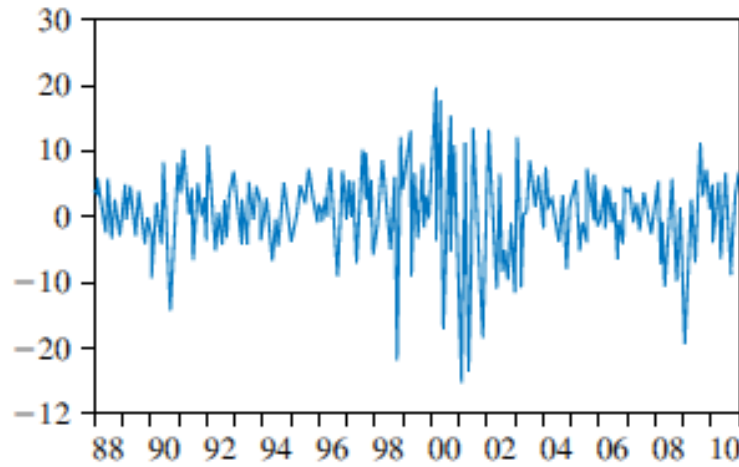


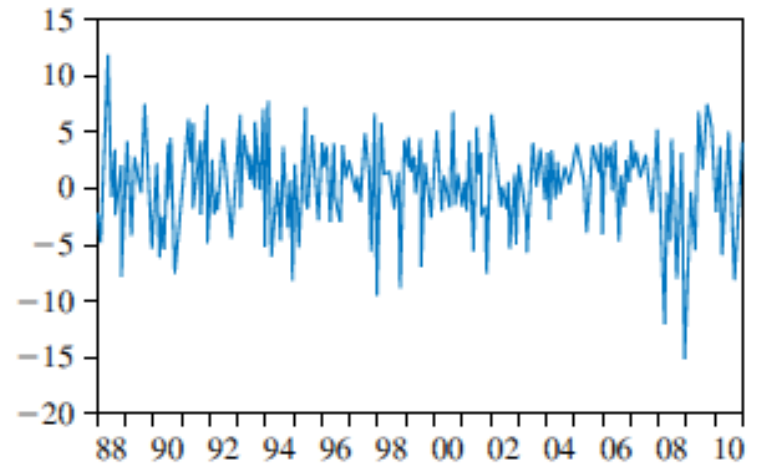
Topic Four:
Time-Varying Volatility and ARCH
Models

- Time-Varying Volatility
- The ARCH Model
- Testing, Estimating, and Forecasting
- Extensions
 - ARCH (q)
 - GARCH
 - T-GARCH
 - GARCH in Mean
 -

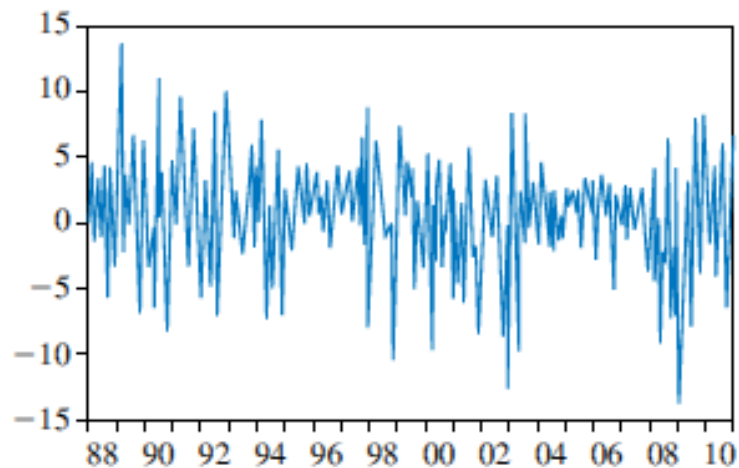
Time-Varying Volatility



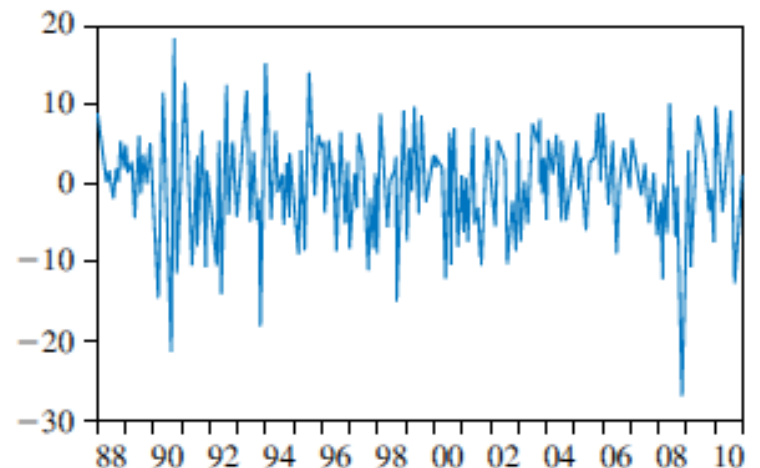
(a) United States: Nasdaq



(b) Australia: All Ordinaries

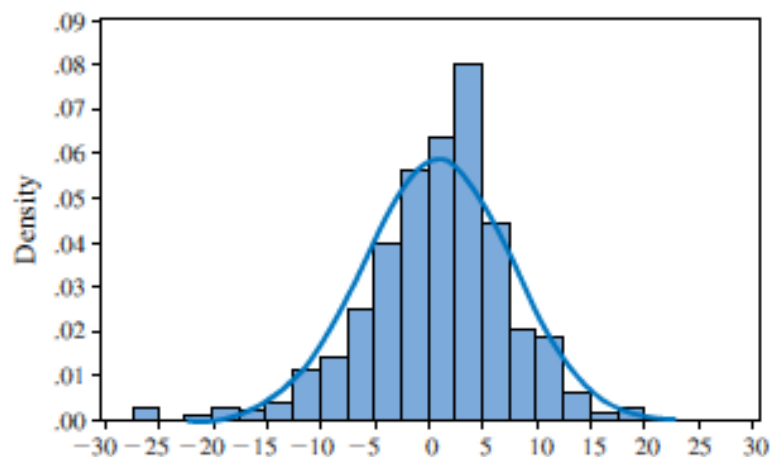


(c) United Kingdom: FTSE

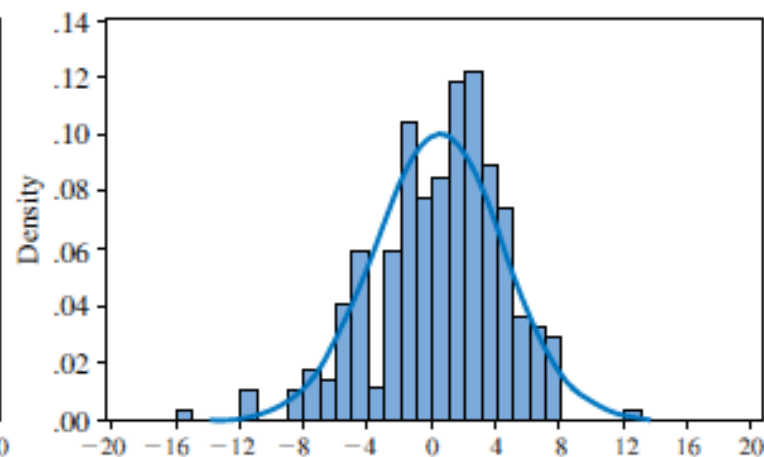


(d) Japan: Nikkei

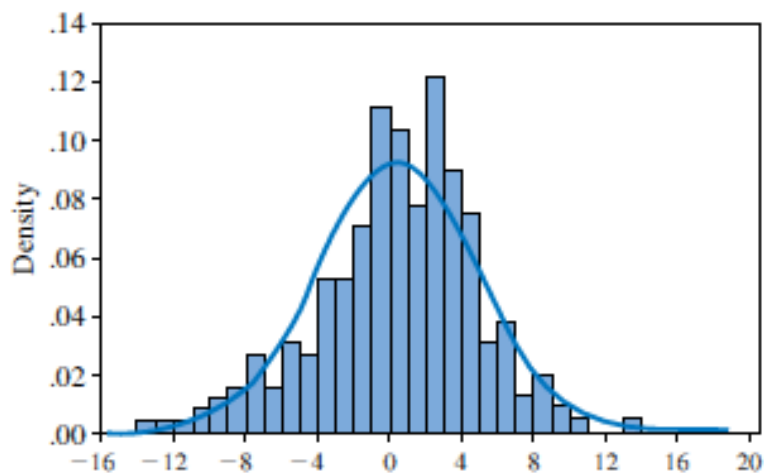
Time-Varying Volatility



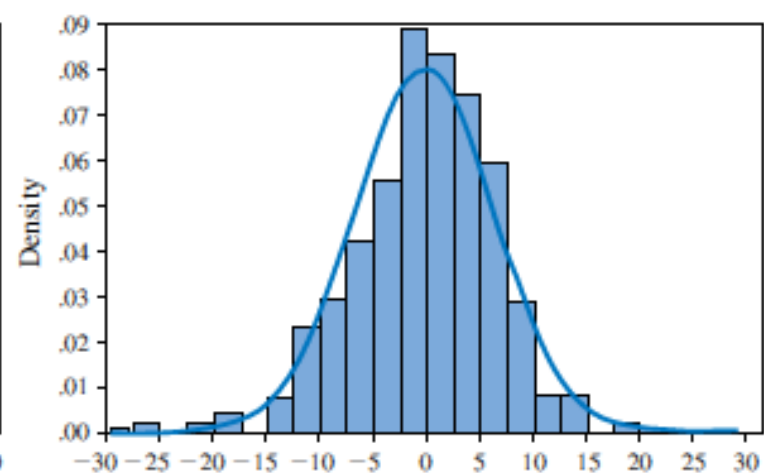
(a) United States: Nasdaq



(b) Australia: All Ordinaries

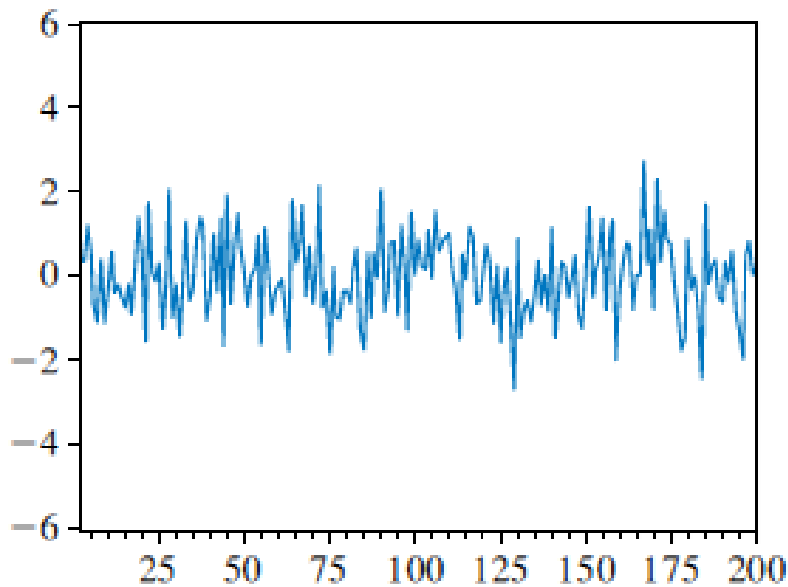


(c) United Kingdom: FTSE

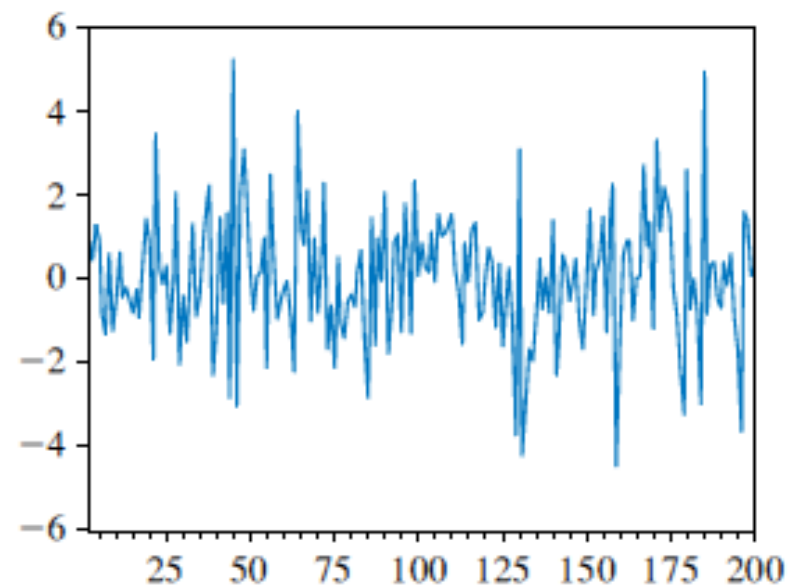


(d) Japan: Nikkei

Time-Varying Volatility



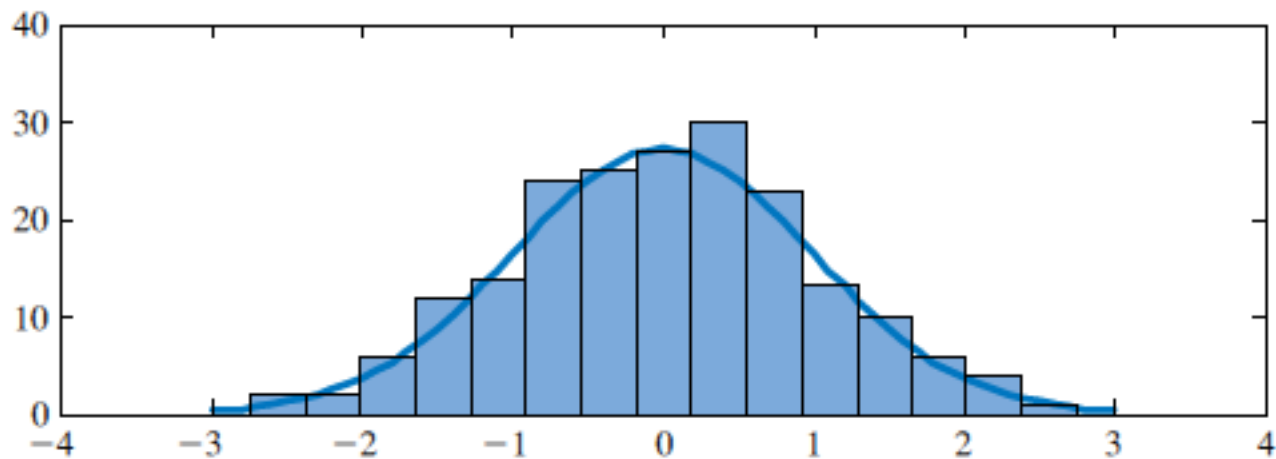
(a) Constant variance: $h_t = 1$



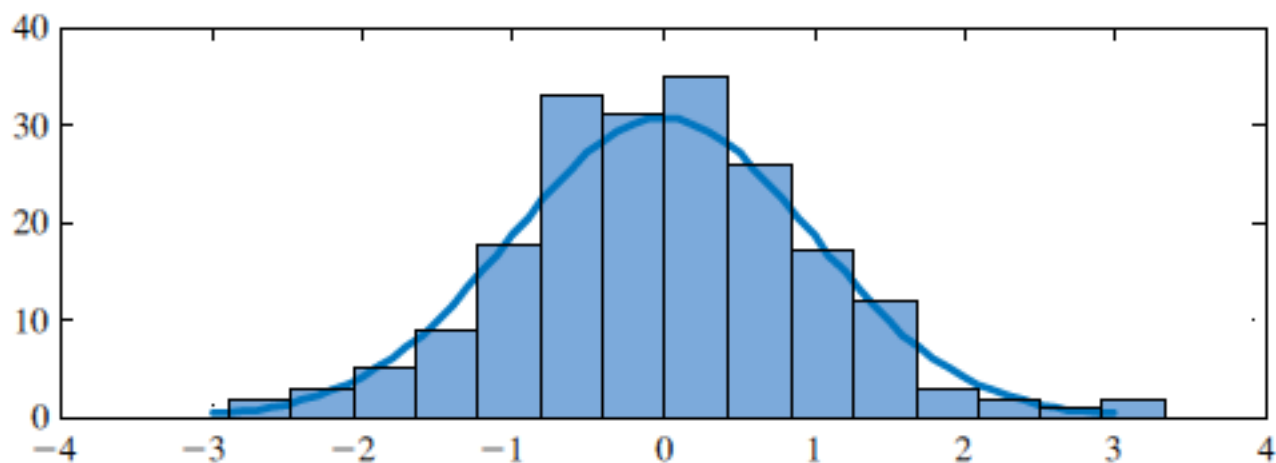
(b) Time-varying variance: $h_t = 1 + 0.8e_{t-1}^2$

Simulated examples of constant and time-varying variances

Time-Varying Volatility



(a) Constant variance



(b) Time-varying variance

The ARCH Model

- Autoregressive conditional heteroskedastic (ARCH) model
- Deals with stationary series, but with conditional variances that change over time

- Consider a model with an AR(1) error term:

$$y_t = \phi + e_t, \quad e_t = \rho e_{t-1} + v_t, \quad |\rho| < 1 \quad v_t \sim N(0, \sigma_v^2)$$

- The **unconditional mean** of the error is:

$$E[e_t] = E[v_t + \rho v_{t-1} + \rho^2 v_{t-2} + \dots] = 0$$

- The **conditional mean** for the error is:

$$E[e_t | I_{t-1}] = E[\rho e_{t-1} | I_{t-1}] + E[v_t] = \rho e_{t-1}$$

The ARCH Model

- The **unconditional variance** of the error is:

$$\begin{aligned} E[e_t - 0]^2 &= E[v_t + \rho v_{t-1} + \rho^2 v_{t-2} + \dots]^2 \\ &= E[v_t^2 + \rho^2 v_{t-1}^2 + \rho^4 v_{t-2}^2 + \dots] \\ &= \sigma_v^2 [1 + \rho^2 + \rho^4 + \dots] \\ &= \frac{\sigma_v^2}{1 - \rho^2} \end{aligned}$$

- The **conditional variance** for the error is:

$$E[(e_t - \rho e_{t-1})^2 | I_{t-1}] = E[v_t^2 | I_{t-1}] = \sigma_v^2$$

The ARCH Model

- Suppose that instead of a conditional mean that changes over time we have a conditional variance that changes over time
- Consider the AR(1) model on slide 7:

$$y_t = \phi + e_t, \quad e_t | I_{t-1} \sim N(0, h_t)$$

$$h_t = \alpha_0 + \alpha_1 e_{t-1}^2, \quad \alpha_0 > 0, \quad 0 \leq \alpha_1 < 1$$

- ARCH: time-varying variances (heteroskedasticity) that depend on (are conditional on) lagged effects (autocorrelation)

The ARCH Model

- The AR(1) model is now an ARCH (1) model

$$y_t = \phi + e_t, \quad e_t | I_{t-1} \sim N(0, h_t)$$

$$h_t = \alpha_0 + \alpha_1 e_{t-1}^2, \quad \alpha_0 > 0, \quad 0 \leq \alpha_1 < 1$$

- The standardized errors are standard normal:

$$\left(\frac{e_t}{\sqrt{h_t}} | I_{t-1} \right) = z \sim N(0, 1)$$

– We can write:

$$E(e_t) = E(z_t) E(\sqrt{\alpha_0 + \alpha_1 e_{t-1}^2}) = 0$$

$$E(e_t^2) = E(z_t^2) E(\alpha_0 + \alpha_1 e_{t-1}^2) = \alpha_0 + \alpha_1 E(e_{t-1}^2)$$

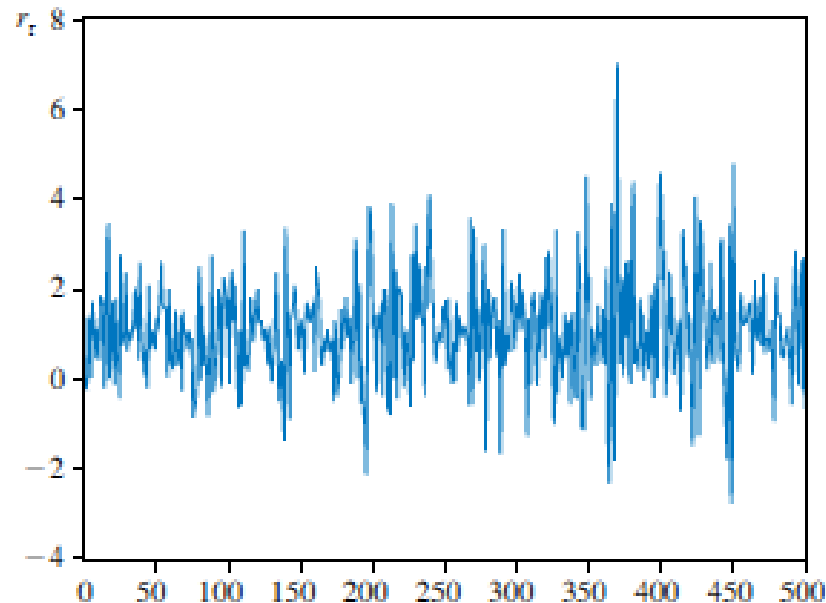
$$\text{var}(e_t) = E(e_t^2) = E(e_{t-1}^2) = \frac{\alpha_0}{1 - \alpha_1}$$

The ARCH Model

- The ARCH model is useful for modeling volatility and especially changes in volatility over time
- The ARCH model is intuitively appealing because it seems sensible to explain volatility as a function of the errors e_t
 - These errors are often called “shocks” or “news” by financial analysts
 - They represent the unexpected
 - According to the ARCH model, the larger the shock, the greater the volatility in the series
 - This model captures volatility clustering, as big changes in e_t are fed into further big changes in h_t via the lagged effect e_{t-1}

Testing, Estimating, and Forecasting

- Testing: Lagrange multiplier (LM) test for ARCH
- Test equation: $\hat{e}_t^2 = \gamma_0 + \gamma_1 \hat{e}_{t-1}^2 + v_t$
- Hypotheses: $H_0 : \gamma_1 = 0$ $H_1 : \gamma_1 \neq 0$
- Example: Shares of Brighten Your Day (BYD) Lighting



Time series of returns for BYD Lighting

- The results for an ARCH test are:

$$\hat{e}_t^2 = 0.908 + 0.353\hat{e}_{t-1}^2 \quad R^2 = 0.124$$

$$(t) \quad (8.409)$$

- The t -statistic suggests a significant first-order coefficient
- The sample size is 500, giving LM test value of $(T - q)R^2 = 62.16$
- Comparing the computed test value to the 5% critical value of a $\chi^2_{(1)}$ distribution ($\chi^2_{(0.95, 1)} = 3.841$) leads to the rejection of the null hypothesis
- The residuals show the presence of ARCH(1) effects.

Testing, Estimating, and Forecasting

- Estimating: maximum likelihood method
- Example: Brighten Your Day (BYD) Lighting
- The estimated models are:

$$\hat{r}_t = \hat{\beta}_0 = 1.063$$

$$\hat{h}_t = \hat{\alpha}_0 + \hat{\alpha}_1 \hat{e}_{t-1}^2 = 0.642 + 0.569 \hat{e}_{t-1}^2$$

$$(t) \qquad (6.877)$$

- The forecast return and volatility are:

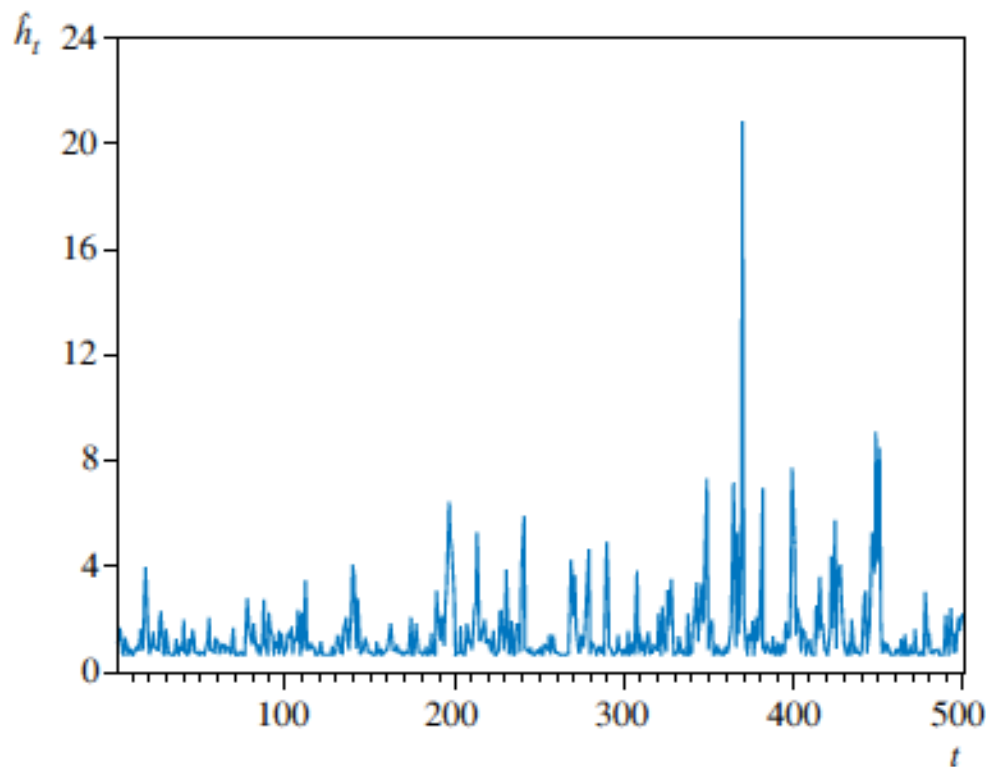
$$\hat{r}_{t+1} = \hat{\beta}_0 = 1.063$$

$$\hat{h}_{t+1} = \hat{\alpha}_0 + \hat{\alpha}_1 \left(r_t - \hat{\beta}_0 \right)^2 = 0.642 + 0.569 (r_t - 1.063)^2$$

- The forecast return and volatility are:

$$\hat{r}_{t+1} = \hat{\beta}_0 = 1.063$$

$$\hat{h}_{t+1} = \hat{\alpha}_0 + \hat{\alpha}_1 (r_t - \hat{\beta}_0)^2 = 0.642 + 0.569(r_t - 1.063)^2$$



- From ARCH(1) to ARCH(q):

- Functional form:

$$h_t = \alpha_0 + \alpha_1 e_{t-1}^2 + \alpha_2 e_{t-2}^2 \dots + \alpha_q e_{t-q}^2$$

- Testing, estimating, and forecasting, are natural extensions of the case with one lag

- One of the shortcomings of an ARCH(q) model is that there are $q + 1$ parameters to estimate, which may lose accuracy in the estimation

- The generalized ARCH model, or GARCH, is an alternative way to capture long lagged effects with fewer parameters

■ From ARCH(q) to GARCH (1, 1) model

■ Re-write $h_t = \alpha_0 + \alpha_1 e_{t-1}^2 + \alpha_2 e_{t-2}^2 \dots + \alpha_q e_{t-q}^2$ as:

$$h_t = \alpha_0 + \alpha_1 e_{t-1}^2 + \beta_1 \alpha_1 e_{t-2}^2 + \beta_1^2 \alpha_1 e_{t-3}^2 + \dots$$

or

$$h_t = (\alpha_0 - \beta_1 \alpha_0) + \alpha_1 e_{t-1}^2 + \beta_1 (\alpha_0 + \alpha_1 e_{t-2}^2 + \beta_1 \alpha_1 e_{t-3}^2 + \dots)$$

but, since:

$$h_{t-1} = \alpha_0 + \alpha_1 e_{t-2}^2 + \beta_1 \alpha_1 e_{t-3}^2 + \beta_1^2 \alpha_1 e_{t-4}^2 + \dots$$

we get: $h_t = \delta + \alpha_1 e_{t-1}^2 + \beta_1 h_{t-1}$

- GARCH (1, 1) model
 - The model is a very popular specification because it fits many data series well
 - It tells us that the volatility changes with lagged shocks (e^2_{t-1}) but there is also momentum in the system working via h_{t-1}
 - One reason why this model is so popular is that it can capture long lags in the shocks with only a few parameters

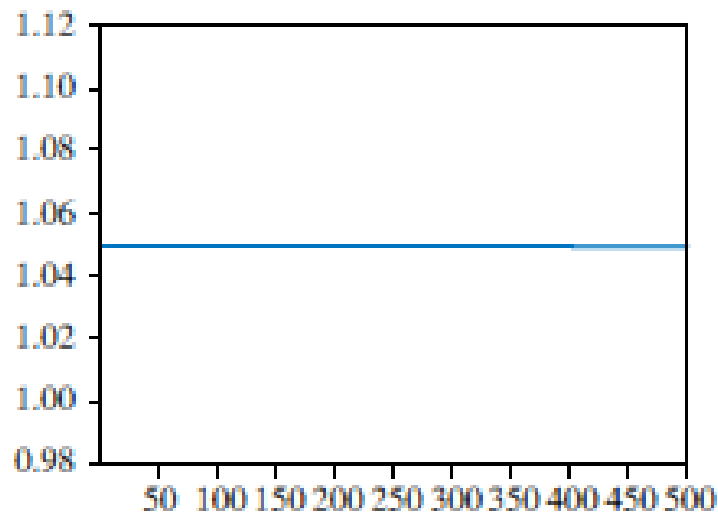
Extensions

- GARCH (1, 1) model
- Example: Brighten Your Day (BYD) Lighting
- The estimated model is:

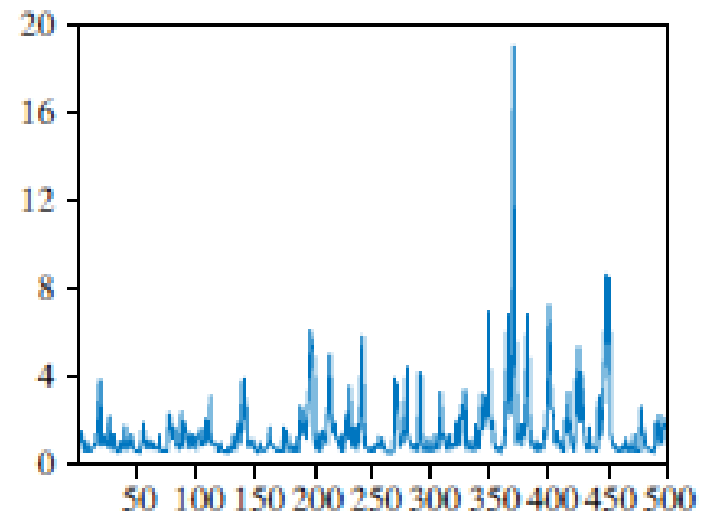
$$\hat{r}_t = 1.049$$

$$\hat{h}_t = 0.401 + 0.492\hat{e}_{t-1}^2 + 0.238\hat{h}_{t-1}$$

(t)
 (4.834)
 (2.136)



(a) GARCH(1,1): $E(r_t) = 1.049$



(b) GARCH(1,1): $h_t = 0.401 + 0.492e_{t-1}^2 + 0.238h_{t-1}$

- The threshold ARCH model, or T-ARCH
- Positive and negative news are treated asymmetrically
- The specification of the conditional variance is:

$$h_t = \delta + \alpha_1 e_{t-1}^2 + \gamma d_{t-1} e_{t-1}^2 + \beta_1 h_{t-1}$$

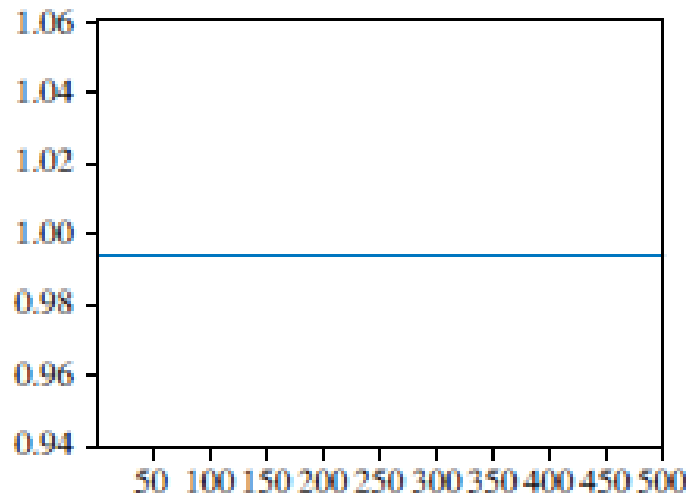
$$d_t = \begin{cases} 1 & e_t < 0 \text{ (bad news)} \\ 0 & e_t \geq 0 \text{ (good news)} \end{cases}$$

Extensions

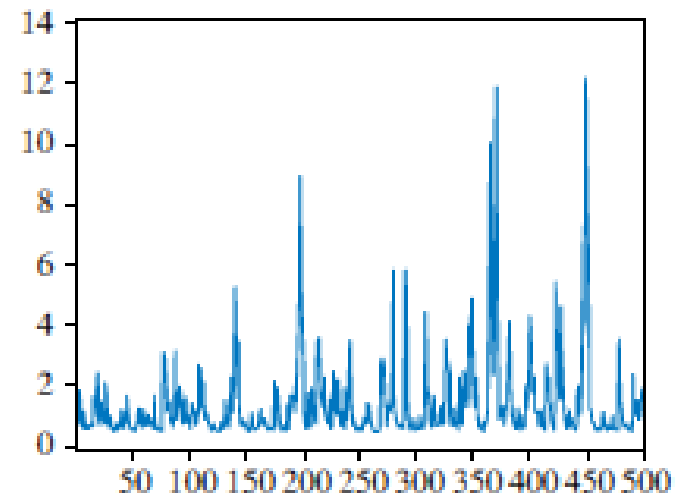
- The threshold ARCH model, or T-ARCH
- Example: Brighten Your Day (BYD) Lighting
- The estimated model is:

$$\hat{r}_t = 0.994 \qquad \hat{h}_t = 0.356 + 0.263\hat{e}_{t-1}^2 + 0.492d_{t-1}\hat{e}_{t-1}^2 + 0.287\hat{h}_{t-1}$$

(t) (3.267) (2.405) (2.488)



(c) T-GARCH(1,1): $E(r_t) = 0.994$



(d) T-GARCH(1,1):
 $h_t = 0.356 + (0.263 + 0.492d_{t-1})e_{t-1}^2 + 0.287h_{t-1}$

- GARCH-in-mean model
- Captures the relationship between risk and return (variance and mean)
- The functional form is:

$$y_t = \beta_0 + \theta h_t + e_t$$

$$e_t \mid I_{t-1} \sim N(0, h_t)$$

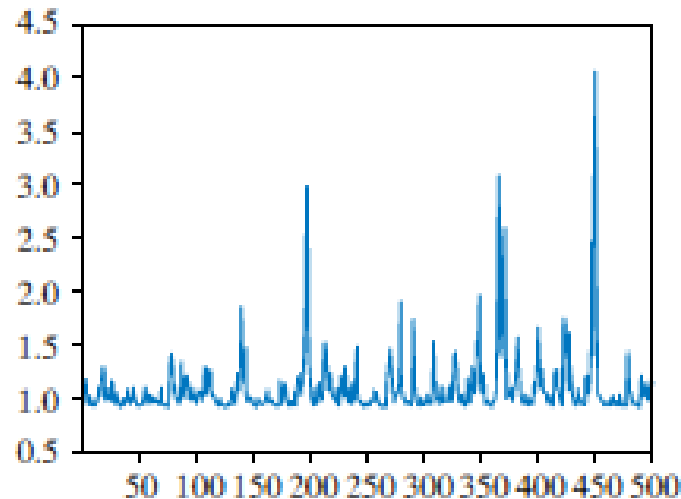
$$h_t = \delta + \alpha_1 e_{t-1}^2 + \beta_1 h_{t-1},$$

$$\delta > 0, \quad 0 \leq \alpha_1 < 1, \quad 0 \leq \beta_1 < 1$$

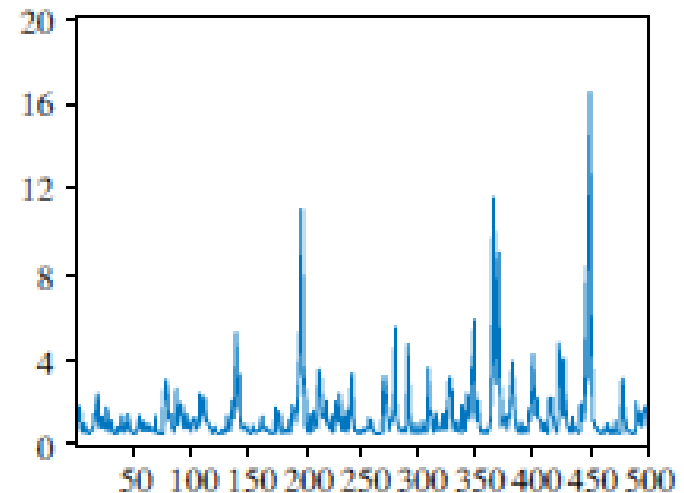
- GARCH-in-mean model
- Example: Brighten Your Day (BYD) Lighting
- The estimated model is:

$$\hat{r}_t = 0.818 + 0.196h_t \quad \hat{h}_t = 0.370 + 0.295\hat{e}_{t-1}^2 + 0.321d_{t-1}\hat{e}_{t-1}^2 + 0.278\hat{h}_{t-1}$$

(t) (2.915) (t) (3.426) (1.979) (2.678)



(e) GARCH-in-mean: $E(r_t) = 0.818 + 0.196h_t$



(f) GARCH-in-mean:
 $h_t = 0.370 + (0.295 + 0.321d_{t-1})\hat{e}_{t-1}^2 + 0.278h_{t-1}$

- Time-Varying Volatility
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